

Customer Complaints and Recovery Effectiveness: A Customer Base Approach

George Knox

Rutger van Oest

George Knox is an Associate Professor at Tilburg University, Warandelaan 2, 5000LE Tilburg, the Netherlands (email G.Knox@tilburguniversity.edu, Tel: 31 13 466 8778).

Rutger van Oest is an Associate Professor at the BI Norwegian Business School, Nydalsveien 37, N-0442 Oslo, Norway (email: rutger.d.oest@bi.no, Tel: 47 4641 0653).

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Abstract

While customer complaints are a well-studied reality of business, no study has measured the impact of actual complaints and recoveries on subsequent customer purchasing. The authors develop a customer base model to investigate how effective recovery is in preventing customer churn. They calibrate it on panel data that track actual purchases, complaints and recoveries for 20,000 new customers of an internet and catalog retailer over 2.5 years. Complaints are associated with a substantial increase in the probability that the customer stops buying, but the size of the increase depends on prior customer experiences: prior purchases mitigate the effect and their impact is long-lasting; prior complaints exacerbate it, but their impact is short-lived. Hence, unless the customer leaves the company after a complaint, or a second failure occurs shortly after the first, the relationship quickly returns to business as usual. Recovery counters the effect of the complaint but in almost all cases does not entirely offset it. The authors use simulation to translate the results to financial impact and discuss implications for researchers and managers.

KEY WORDS: *Complaints, Complaint Management, Churn, Failure, Recovery*

Companies need to maintain long-run relationships with their customers in order to survive. Customer base analysis aids managers by, among other things, forecasting future purchasing and identifying which customers are most at risk of leaving (Fader and Hardie 2005; Gupta et al. 2006). The vast majority of customer base models rely exclusively on customers' prior transaction history to make these predictions (e.g., Fader, Hardie and Lee 2005a, b; Schmittlein, Morrison and Colombo 1987), though more recent work incorporates non-transactional elements like customer satisfaction (Ho, Park and Zhou 2006) and service quality (Braun, Schweidel and Stein 2013).

Customer complaints are important non-transactional events, because they represent critical turning points in the company's relationship with its customers (Blattberg, Kim and Neslin 2008). They are an everyday reality of business in virtually any industry and are typically stored in data warehouses, making them easily accessible to managers. How complaints are handled (i.e., the recovery) marks the "acid test of a company's customer orientation" (Homburg and Fürst 2005) and may have far-reaching positive or negative consequences for the company's relationship with its customers. An extraordinary recovery may turn complaining customers into loyal ones and generate more goodwill than if the failure had not occurred in the first place (Hart, Heskett and Sasser 1990); an ineffective recovery can leave customers feeling even worse about the company and drive them to leave (Bitner, Booms and Tetreault 1990). If recovery is a strategic resource (Hart, Heskett and Sasser 1990), companies need to know how effective it is in driving subsequent customer behavior.

The goal of our study is to incorporate these critical non-transactional elements—complaints and recoveries—into customer base analysis and investigate their effects on customer churn, i.e., the likelihood that the customer permanently stops buying from the company. In

transactional (non-contractual) settings—applicable to many retailers and service providers—this poses a challenge, because churn is not directly observed and hence must be inferred using observed customer activity (Kumar and Reinartz 2012).

Panel data that track actual customer purchases, complaints and recoveries over time have not been used before to study complaint management (Rust and Chung 2006, p. 563) and are critical to our objective. Van Oest and Knox (2011) use actual purchases and complaints to forecast future purchasing, but ignore the role of recovery. Several other studies use actual purchase—but not complaint and recovery—data to study the effects of customer satisfaction (e.g., Bolton 1998; Cooil et al. 2007), reported satisfaction with recovery (Gilly and Gelb 1982), extreme experiences (Bolton, Lemon and Bramlett 2006), and product returns (Reinartz and Kumar 2003) on future purchasing. Virtually all service recovery studies take the customer's perspective (Parasuraman 2006) and use survey-based approaches, sometimes combined with experimental data (e.g., Smith and Bolton 1998; Smith, Bolton and Wagner 1999), to investigate the consequences for customer perceptions and intentions. While crucial for understanding the customer's process, these survey-based metrics have limited managerial relevance (Davidow 2003; Mittal and Kamakura 2001), in part because behavioral intentions may be systematically biased and the link to actual behavior may be weak (Gupta and Zeithaml 2006; Chandon, Morwitz and Reinartz 2005). Therefore, “the most fertile area for research on customer impact pertains to how customer behavior (rather than attitudes or intentions) responds to changes in marketing actions” (Rust et al. 2004, p. 84).

Our customer base approach is novel, because we take the company's perspective and examine the consequences of actual complaints and recoveries on actual purchasing. We test whether three fundamental hypotheses from the service recovery literature hold with actual

customer behavior and we test a new hypothesis about how long the effects of prior purchases and complaints last in the customer relationship; such a hypothesis would be difficult to test without panel data. We control for unobserved heterogeneity in customer response and endogeneity in the company's recovery decision to separate out effects due to the recovery from those due to unobserved customer characteristics or company targeting. Our model can help managers decide whether to provide recovery based on balancing its costs and benefits, an important yet largely ignored issue (Parasuraman 2006).

We contribute several new findings to the collective knowledge on customer complaints. Complaints are associated with a substantial increase in churn, but the size of the increase depends on prior customer experiences: prior purchases mitigate the effect and have a long-lasting multi-year impact; prior complaints exacerbate it, but their impact is short-lived because of the changing composition of the customer base—the most dissatisfied customers stop buying and hence drop out of the customer base, leaving only customers who are willing to give the company another chance. So, unless the customer leaves after a complaint, or a second failure follows closely after the first, the relationship quickly returns to business-as-usual. Finally, though recovery counters the effect of a complaint and helps reset the relationship, in almost all cases it does not entirely offset the complaint; we find little evidence for a recovery paradox.

In the next sections, we present the conceptual framework and develop the hypotheses underlying our empirical analysis. Next, we describe the data set and specify the model. We then discuss the empirical findings, the results of the hypothesis tests, and the financial consequences of complaints and recoveries. We conclude with the implications of our study to researchers and managers.

Conceptual Framework

Future purchasing depends on the likelihood of churn following two types of events—complaints and purchases—where churn refers to the event that the customer permanently becomes inactive, i.e., stops buying from the company forever. Our main interest centers on explaining churn following a complaint, so we formulate our hypotheses using this as our dependent variable. Nevertheless, we also control for churn following a purchase and its drivers. Figure 1 presents the framework.

[Insert Figure 1 About Here]

We define complaints as customer-initiated expressions of dissatisfaction to the company (e.g., Landon 1980); they represent an opportunity for the company to remedy a problem (Bijmolt et al. 2010). Recovery refers to the set of actions a company takes in response to the failure in order to keep the customer's goodwill (Grönroos 1988). The service recovery literature typically views recovery from the customer's perspective by using customer satisfaction with complaint handling (e.g., Smith and Bolton 1998; Tax, Brown and Chandrashekar 1998). In contrast, we approach it from the company's perspective as a dichotomous outcome: the company recovers the complaint if it makes an attempt that is not refused by the customer. Hence, no recovery is provided if the company does not make any (further) attempt to resolve the problem; the no-recovery outcome includes cases where the customer refused the proposed solution and the company did not make a new offer. We characterize prior complaints and purchases by their incidence and timing (Van Oest and Knox 2011). Similar to Ansari, Mela and Neslin (2008) and Van Diepen, Donkers and Franses (2009), we weigh these prior customer experiences by their recency to capture the notion that events are less salient (and therefore likely

discounted more) the longer ago they occurred. Our last hypothesis concerns the speed of decay, i.e., the degree of discounting, for prior complaints relative to prior purchases.

Impact of Prior Complaints on Churn after Complaint

One complaint may signal a forgivable error, but voicing more than one complaint may indicate a more systemic problem. Attribution theory predicts that the customer will react more negatively if the cause of the complaint is perceived to be permanent rather than temporary (Folkes 1984; Maxham and Netemeyer 2002). The first failure decreases trust but also creates uncertainty as to whether the lower level of trust is justified; a second failure acts as confirmation and decreases both level of trust and degree of uncertainty (Chandrashekar et al. 2007; Rotte et al. 2006). Memory research suggests that a second failure may reactivate the effects of the first failure if recent enough (Solomon 2006); this reinforcement effect (Van Oest, Van Heerde and Dekimpe 2010) makes the impact of the second failure more severe. If there are prior incidents, customers are less satisfied, have lower self-stated repurchase intentions (Maxham and Netemeyer 2002; Van Doorn and Verhoef 2008), and hence are more likely to leave.

H1: Prior complaints increase the likelihood of churn following a complaint.

Impact of Prior Purchases on Churn after Complaint

High prior satisfaction with the company relationship may influence customer responsiveness to failure in opposite ways. Accumulated goodwill can provide a buffer and make customers less responsive when a failure occurs. However, positive prior experience can also lead to high expectations which are harder to meet and may therefore trigger more negative customer response (Bolton 1998; Tax, Brown and Chandrashekar 1998). There is considerable

survey evidence that the goodwill (or buffer) effect of prior satisfaction dominates the expectations effect, i.e., customers become less responsive to failure when satisfaction increases (e.g., Anderson and Sullivan 1993; Tax, Brown and Chandrashekar 1998; Van Doorn and Verhoef 2008). In line with the goodwill explanation, a failure is inconsistent with prior experience for a satisfied customer and hence this customer will keep a lower level of trust in the company with large uncertainty; this uncertainty limits the negative impact of the failure (Chandrashekar et al. 2007; Rotte et al. 2006). A lengthy and extensive purchase history indicates a stable and satisfactory customer-company relationship (Cooil et al. 2007; Gupta and Zeithaml 2006; Rust and Chung 2006) and hence should make the customer more forgiving when a complaint occurs.

H2: Prior purchases decrease the likelihood of churn following a complaint.

Impact of Recovery on Churn after Complaint

The customer starts from a deficit position when a complaint occurs (Andreassen 1999; Oliver 1997) and expects the company to resolve the issue. Customers have expectations about the quality of service, but also about the quality of the company's recovery following a service failure (McCollough, Berry and Yadav 2000). They encode a recovery outcome as a gain or loss depending on their expectations. A good recovery (positive disconfirmation) is a perceived gain; no or insufficient recovery (negative disconfirmation) implies a perceived loss. While the company may not be able to avoid the failure itself, it has control over the recovery provided (Hart, Heskett and Sasser 1990). The gain-loss outcome of recovery and the fact that the company has control over it should make customers responsive to what is received (Folkes 1984; Maxham 2001).

Prospect theory says that customers are more sensitive to losses than gains; worse-than-expected service is more harmful than better-than-expected service is beneficial (e.g., Anderson and Sullivan 1993; Rust et al. 1999). While no or insufficient recovery (a loss) may have large negative consequences for retention, the positive effect of the size of a good recovery (a gain) is limited (Davidow 2003). Hence, exceeding the threshold is the primary objective of recovery; the precise amount is secondary.

If the recovery is worse than expected, i.e., below the threshold, there is a double deviation from expectations due to both the service failure and the company's unsuccessful recovery (Bitner, Booms and Tetreault 1990). This double deviation results in low perceived justice and low satisfaction (e.g., Maxham and Netemeyer 2003) and decreases self-reported customer loyalty (Andreassen 1999; Orsingher, Valentini and De Angelis 2010). On the other hand, if the recovery is better than expected, the negative effect of the failure may be partially or completely mitigated.

In some cases, customers may respond so well that their satisfaction or self-stated repurchase levels are higher than they would have been had no failure occurred (De Matos, Henrique and Rossi 2007; Parasuraman 2006). While some studies offer evidence for this recovery paradox (e.g., Hart, Heskett and Sasser 1990; Maxham and Netemeyer 2002; Smith and Bolton 1998), others report that recovery effectiveness may offset but does not exceed the failure's impact (Andreassen 2001) or does not entirely offset it (Maxham 2001; McCollough, Berry and Yadav 2000). Importantly, none of these studies use actual purchase, complaint and recovery data to estimate recovery effectiveness.

Most extant studies confirm that recovery matters (e.g., Gilly and Gelb 1982; Smith and Bolton 1998), with a few studies reporting either no significant effect (Bolton 1998; Van Doorn

and Verhoef 2008) or that it depends on the attribution context (Grewal, Roggeveen and Tsiros 2008). Taking everything together,

H3: Recovery decreases the likelihood of churn following a complaint.

Impact of Prior Complaints and Purchases on Churn after Purchase

Prior complaints and purchases may also drive churn in the absence of a complaint; though they are not the focus of our study, we control for them. We expect that prior complaints increase the likelihood of churn after a purchase, because they are negatively related to satisfaction and hence purchase intentions (Bolton 1998; Gupta and Zeithaml 2006; Gustafsson, Johnson and Roos 2005); most studies have found that even a better-than-expected recovery does not fully mitigate the negative effects of the underlying failure. In contrast, many prior purchases should reduce the likelihood of churn after a purchase, as they signal satisfaction with the company (e.g., Cooil et al. 2007; Gupta and Zeithaml 2006) and many prior experiences increase trust and decrease perceived risk of future purchasing (Rust et al. 1999).

Speed of Decay in Impact of Prior Complaints Relative to Prior Purchases

We provide two competing hypotheses from different perspectives. The underlying mechanisms lead to opposite predictions; the dominating mechanism determines which hypothesis (if any) is confirmed.

The first perspective is that the impact of negative experiences lasts longer than that of positive experiences (Baumeister et al. 2001); negative events tend to occur relatively infrequently but often with large consequences, making them both accessible in memory and diagnostic (Feldman and Lynch 1988; Menon, Raghurir and Schwarz 1995). This can manifest

itself in multiple ways: negative emotions trigger more cognitive processing and stronger changes in behavior than positive emotions (e.g., Schwarz 1990); negative events also affect relationships more than positive ones (Levenson and Gottman 1983). Hence, the impact of low-frequency and mostly negative emotional events like complaints should last longer (i.e., decay more slowly) than the impact of high-frequency mostly positive events like purchases (Fiske 1980; Nisbett and Ross 1980). This leads to the first competing hypothesis:

H4a: The effect of prior complaints on churn lasts longer than that of prior purchases.

The second perspective amounts to a potentially strong sorting effect (e.g., Fader and Hardie 2010), where the most dissatisfied complaining customers drop out of the customer base and hence do not contribute to the rate of decay of prior complaints. The customers who remain with the company and (by definition) give it another chance are the ones on which the rate of decay is estimated. Within this group, the effect of the complaint on the future likelihood of churn is likely short-lived; while the complaint may remain accessible in memory, it quickly becomes less diagnostic once one has decided to stay with the company after the complaint. These customers may be dissatisfied and suffer in silence, but remain loyal in terms of future purchasing (Hirschman 1970).

H4b: The effect of prior complaints on churn lasts shorter than that of prior purchases.

Data

Our transaction and complaint data are from a major U.S. internet and catalog retailer, with over 2,000 employees and ranked in the Top 50 largest direct marketers by *Catalog Age* and *Internet Retailer*. We observe 20,000 new customers from their first purchase with the company,

in one of the first 10 weeks of 2001, until June 30, 2003, i.e., at most 2.5 years since their initial purchase. Customers made on average 2.3 purchases (ranging from 1 to 26) with an average (completed) inter-purchase time of 192 days; they spent on average \$57.32 per order. The correlation between inter-purchase time and spending per purchase is nearly zero ($p = .64$).

The novel feature of our data is the complaint log file, which identifies the customer, date, and a description of the complaint incident. There are 922 complaints recorded from 805 customers (4% of total customers). Most complaints are voiced soon after a purchase: 18% of all complaints occur on the same day as a purchase and 80% occur within three weeks; the average time to voice a complaint is 28 days. The across-customer correlation between number of purchases and number of complaints is .14 ($p < .001$). We hired two MPhil students to independently code whether the company provided recovery for each of the 922 complaint events. This resulted in 884 recovered complaints and 38 cases without recovery, with a high intercoder reliability of .973 (Perreault and Leigh 1989). Coding discrepancies were resolved by a third judge (one of the authors).

A Model for Purchasing, Complaining and Churn

We assume that customers go through two stages in their latent lifetime with the company (e.g., Schmittlein, Morrison and Colombo 1987). All customers start in an active state and remain there until they become permanently inactive, when churn takes place. Since our setting is non-contractual (i.e., transactional), the time at which churn occurs is not observed and must be inferred from observed patterns in purchasing (Fader, Hardie and Lee 2005a, b; Schmittlein, Morrison and Colombo 1987) and complaining behavior (Van Oest and Knox 2011). Similar to

the BG/NBD (Fader, Hardie and Lee 2005a), we assume that churn can only occur after an event, i.e., a purchase or a complaint.

While a customer must be active to either purchase or complain, the absence of these events over time has two possible explanations: the customer permanently became inactive after the last event and will never purchase or complain again, or the customer is still active and the next event will occur beyond the observation horizon. For a frequent customer, a long hiatus in purchasing and complaining is a strong signal of churn; for an infrequent customer, a long hiatus is a weaker signal, because it is consistent with both active (but infrequent) and inactive states.

The flow chart in Figure 2 describes the customer's sequence of steps from the moment of first purchase (the purchase timing block, initialized at value 0) until the moment of churn when the customer permanently moves to the absorbing inactive state, i.e., leaves the company. After a purchase, churn occurs with probability $\pi_{\text{churn_pur}}$ and the customer remains active with probability $1 - \pi_{\text{churn_pur}}$. Next, if still active, the customer becomes unhappy and voices a complaint with probability π_{com} , with the complementary probability of no complaint being $1 - \pi_{\text{com}}$. If a complaint is voiced, it either occurs on the same day as a purchase with probability π_{sameday} , or is delayed with probability $1 - \pi_{\text{sameday}}$, where the number of days of delay follows a timing distribution. The customer leaves the company after the complaint with churn probability $\pi_{\text{churn_com}}$, or remains active with probability $1 - \pi_{\text{churn_com}}$. If the customer remains active, she may still be unhappy and voice another complaint with probability π_{another} , with the timing since last complaint again being drawn from a timing distribution, or not voice any further complaints with probability $1 - \pi_{\text{another}}$.

[Insert Figure 2 About Here]

The process above describes an iteration of one purchase and possibly multiple complaints. Next, a new iteration starts for an active customer, where the time of the next purchase relative to the last event (purchase or complaint) follows a timing distribution. Iterations continue until the customer churns.

To test our hypotheses, we make the $\pi_{\text{churn_com}}$ in Figure 2 a function of the customer's prior complaints, prior purchases, and whether the company provides recovery. Because churn is unobserved, we need to infer it from the timing of purchases and complaints; to accomplish this we need to specify their corresponding timing distributions. We next describe these model components; we use the index $i = 1, \dots, N$ for customers and the index $j = 1, \dots, J_i$ for customer i 's events, i.e., purchases and complaints.

Event Timing

For $j = 2, \dots, J_i$ (time is initialized at zero for event $j = 1$), we capture *purchase* timing relative to the last event by a Weibull density. The constant shape parameter is α_{pur} and we allow the scale parameter $\tilde{\beta}_{\text{pur},i,j}$ to depend on the last event's type (Park and Fader 2004), i.e., a purchase or a complaint. The probability density is

$$(1) \quad f_{\text{pur}}(t_{i,j}) = \alpha_{\text{pur}} \tilde{\beta}_{\text{pur},i,j}^{-\alpha_{\text{pur}}} t_{i,j}^{\alpha_{\text{pur}}-1} \exp(-[t_{i,j}/\tilde{\beta}_{\text{pur},i,j}]^{\alpha_{\text{pur}}}),$$

with

$$(2) \quad \tilde{\beta}_{\text{pur},i,j} = \exp(\beta_{\text{pur},0} + \beta_{\text{pur},1} \text{COM}_{i,j-1}),$$

where $\text{COM}_{i,j-1} = 1$ if the last event $j - 1$ was a complaint and $\text{COM}_{i,j-1} = 0$ if it was a purchase. The Weibull density in (1) is flexible and allows for non-monotonic shapes. If $\alpha_{\text{pur}} = 1$, it collapses into the more restrictive and monotonically decreasing exponential density that is often used (Fader, Hardie and Lee 2005a, b; Schmittlein, Morrison and Colombo 1987).

Similar to (1), we assume that *complaint* timing for $j = 2, \dots, J_i$ is given by another Weibull density with its scale parameter depending on $\text{COM}_{i,j-1}$:

$$(3) \quad f_{\text{com}}(t_{i,j}) = \alpha_{\text{com}} \tilde{\beta}_{\text{com},i,j}^{-\alpha_{\text{com}}} t_{i,j}^{\alpha_{\text{com}}-1} \exp(-[t_{i,j}/\tilde{\beta}_{\text{com},i,j}]^{\alpha_{\text{com}}}),$$

with

$$(4) \quad \tilde{\beta}_{\text{com},i,j} = \exp(\beta_{\text{com},0} + \beta_{\text{com},1} \text{COM}_{i,j-1}).$$

Customer Churn

In accordance with H1-H3, the probability that the customer churns after a *complaint*, $\pi_{\text{churn_com},i,j}$, may depend on the customer's prior complaints and purchases and whether the current complaint is recovered by the company¹:

(5)

$$\pi_{\text{churn_com},i,j} = \frac{\exp(\delta_{\text{com},0} + \delta_{\text{com},1} \text{PCOM}_{i,j} + \delta_{\text{com},2} \text{PPUR}_{i,j} + \delta_{\text{com},3} \text{REC}_{i,j})}{1 + \exp(\delta_{\text{com},0} + \delta_{\text{com},1} \text{PCOM}_{i,j} + \delta_{\text{com},2} \text{PPUR}_{i,j} + \delta_{\text{com},3} \text{REC}_{i,j})},$$

where $\text{REC}_{i,j}$ is the 0/1 indicator variable for recovery. The variables $\text{PCOM}_{i,j}$ and $\text{PPUR}_{i,j}$ capture customer i 's prior complaints and purchases by the time of event j . If there is no recency weighting, $\text{PCOM}_{i,j} = \sum_{k=1}^{j-1} \text{COM}_{i,k}$, i.e., the number of prior complaints from initial event 1 to the most recent event $j - 1$, where the 0/1 indicator variable $\text{COM}_{i,k}$ ensures that only complaint events are counted. Similarly, $\text{PPUR}_{i,j} = \sum_{k=1}^{j-1} (1 - \text{COM}_{i,k})$ counts the number of prior purchases, where the 0/1 indicator variable $1 - \text{COM}_{i,k}$ ensures that only purchase events (not complaints) are included. We incorporate recency weighting using an exponential decay pattern (Ansari, Mela and Neslin 2008; Van Diepen, Donkers and Franses 2009), so that the final forms of the two variables are:

(6)

$$\begin{aligned} \text{PCOM}_{i,j} &= \sum_{k=1}^{j-1} \lambda_{\text{com}}^{T_{i,k \rightarrow j}} \text{COM}_{i,k}, \\ \text{PPUR}_{i,j} &= \sum_{k=1}^{j-1} \lambda_{\text{pur}}^{T_{i,k \rightarrow j}} (1 - \text{COM}_{i,k}), \end{aligned}$$

where $T_{i,k \rightarrow j}$ is the number of days between customer i 's prior event k and current event j , and $0 < \lambda_{\text{com}} \leq 1$ and $0 < \lambda_{\text{pur}} \leq 1$ are the decay parameters for prior complaints and purchases, respectively. H4a (H4b) corresponds to $\lambda_{\text{com}} > (<) \lambda_{\text{pur}}$. The summation index k in (6) runs over all event types (both purchases and complaints), where the 0/1 indicator variables ensure that only the appropriate event type is included. If $\lambda_{\text{com}} = \lambda_{\text{pur}} = 1$, there is no recency weighting and prior complaints and purchases would become unweighted counts.

We let the customer's likelihood of churn after a *purchase*, $\pi_{\text{churn_pur},i,j}$, vary according to prior complaints and purchases in a similar way as (5):

(7)

$$\pi_{\text{churn_pur},i,j} = \frac{\exp(\delta_{\text{pur},0} + \delta_{\text{pur},1} \text{PCOM}_{i,j} + \delta_{\text{pur},2} \text{PPUR}_{i,j})}{1 + \exp(\delta_{\text{pur},0} + \delta_{\text{pur},1} \text{PCOM}_{i,j} + \delta_{\text{pur},2} \text{PPUR}_{i,j})}.$$

Unobserved Customer Heterogeneity

We capture unobserved heterogeneity using latent classes. We make all parameters in the customer's timing and churn decisions (1)-(5) and (7) class-specific as well as the customer's probability of voicing a complaint, i.e., π_{com} . We keep the probabilities π_{sameday} and π_{another} homogeneous; they are necessary for predicting future purchasing, but not the focus of our hypotheses and are based on few observations.

Endogeneity of the Company's Decision of Whether to Provide Recovery

Whether the company provides a recovery to a complaining customer may be partially driven by observed summaries of customer behavior, such as prior complaints and purchases. Other unobservable factors such as attributes of the customer (e.g., irate, friendly) or the complaint (e.g., attribution, severity) may also play a role. Hence, the recovery decision is not entirely random; ignoring the association between recovery and the customer's propensity to purchase, complain and leave the company may lead to biased parameter estimates. We establish an association by estimating the parameters governing the customer's decisions and the company's recovery decision simultaneously in the latent class framework (e.g., Donkers et al. 2006; Schweidel and Knox 2013). We let the probability that the company provides recovery depend on the customer's prior complaints and purchases:

(8)

$$\pi_{\text{rec},i,j} = \frac{\exp(\delta_{\text{rec},0} + \delta_{\text{rec},1}\text{PCOM}_{i,j} + \delta_{\text{rec},2}\text{PPUR}_{i,j})}{1 + \exp(\delta_{\text{rec},0} + \delta_{\text{rec},1}\text{PCOM}_{i,j} + \delta_{\text{rec},2}\text{PPUR}_{i,j})},$$

where $\text{PCOM}_{i,j}$ and $\text{PPUR}_{i,j}$ are defined by (6) and the parameters $\delta_{\text{rec},0}$ and $\delta_{\text{rec},2}$ are class-specific. We keep $\delta_{\text{rec},1}$ fixed across classes, because it is hard to identify empirically; relatively few customers have prior complaints (limited variation in the independent variable) and complaints are recovered in a vast majority of cases (also limited variation in the dependent variable). We estimate all model parameters in (1)-(8) across all classes jointly using maximum likelihood. Appendix A provides the technical details.

Results

To determine the number of latent classes, we start with one and increase this number until the BIC criterion does not decrease anymore (e.g., Donkers et al. 2006). Using the whole observation period, the BIC score is lowest for two classes; this result does not change when a shorter calibration period is taken. When we randomly divide customers in our sample into two equally sized groups of 10,000 and fit the two-class model separately to these two sub-samples, we again find an optimal two-class solution with similar relative class sizes and class-specific estimates. Hence, our latent class results appear to be stable.

We validate our model by comparing its out-of-sample predictions to those of two benchmark models that ignore complaints and recoveries: a two-class nested “purchase-only” model and the well-known BG/NBD model (Fader, Hardie and Lee 2005a). We find that our out-of-sample predictive performance is better than both benchmark models. The results of the model validation exercise are in Appendix B. We next report the parameter estimates and their implications for our hypotheses about customer churn. Then we quantify our estimates further by simulating the financial consequences of complaints and recoveries.

Parameter Estimates and Hypothesis Tests

We use the full 130-week sample to calibrate the model and test our hypotheses. Table 1 presents all parameter estimates and standard errors specific to each latent class. Class 1 comprises 82% of the customer base; Class 2 represents the remaining 18%.

[Insert Table 1 About Here]

The baseline propensity to churn following a complaint ($\delta_{\text{com},0}$) is larger in Class 1 than in Class 2. According to H1, more prior complaints increase the likelihood of churn after a complaint; H2 says that more prior purchases decrease the likelihood of churn after a complaint. In Class 1, we find marginal support for H1 ($\delta_{\text{com},1} = 2.70$, t-statistic = 1.82) and strong support for H2 ($\delta_{\text{com},2} = -1.38$, t-statistic = -4.17). In Class 2, we also find strong support for H2 ($\delta_{\text{com},2} = -0.49$, t-statistic = -3.30), but not for H1: the estimate of prior complaints on churn after a complaint is negative and has a very large standard error. This parameter may not be well identified in Class 2, because the prior complaints variable PCOM is usually very close to zero when complaints occur in this class.

Consistent with H3, recovery leads to a significant drop in the likelihood of churn following a complaint in Class 1 ($\delta_{\text{com},3} = -1.13$, t-statistic = -2.03). For instance, for an average customer in our database, the likelihood of churn drops from 67% to 40% if the company provides recovery to the complaint. In the smaller Class 2, the effect of recovery is as expected, i.e., negative, but not significant. In sum, five out of six coefficients for H1-H3 in the two classes are as anticipated; all three coefficients in the large Class 1 are (marginally) significant and one of the two well-identified coefficients in Class 2 is significant at the 1% level.

The baseline propensity to leave the company following a purchase is lower than a complaint for both classes. In Class 1, customer churn following a purchase increases with prior complaints ($\delta_{\text{pur},1} = 2.41$, t-statistic = 2.38) and decreases with prior purchases ($\delta_{\text{pur},2} = -1.12$, t-statistic = -19.23); the effects are smaller in magnitude but in the same direction for Class 2.

Taken together, what do these estimates imply about the extent of recovery effectiveness: can the probability of churn after a recovered complaint be less than after a hassle-free purchase? For a customer with one recent purchase and zero prior complaints (HPUR = 1, HCOM = 0),

offering recovery reduces the probability of churn from 72% to 46%, but the likelihood of churn after a hassle-free purchase is still lower, i.e., 21%. While this pattern holds for the vast majority of cases in our data, the model does not completely rule out the paradox. Using our estimates, we can show that churn is more likely after a hassle-free purchase than a recovered complaint when the prior purchases variable PPUR is larger than 5.5.² Since this threshold is extremely high relative to the empirical range found in our data—only 15 (1.6%) out of 922 complaints—we can conclude that the conditions under which the paradox occurs are unlikely.

Turning to the recovery decision, we note that the baseline propensity for the company to provide recovery is high, because a vast majority of complaints are recovered. Customers with more (and more recent) prior complaints are less likely to receive a recovery ($\delta_{\text{rec},1} = -3.20$, t-statistic = -2.36).³ Customers with more prior purchases are slightly more likely to receive a recovery in both classes, but the coefficient is not significant.

Do the effects of complaints last longer or shorter than purchases (H4)? The decay parameter of complaints $\lambda_{\text{com}} = .9121$ is significantly lower than the decay parameter of purchases $\lambda_{\text{pur}} = .9978$ ($\chi^2_1 = 15.98, p < .01$), in support of H4b. One way to quantify how long an effect lasts is its halving time—the number of days it takes for the prior purchases or prior complaints variable to lose half of its value. The halving time is 315 days for purchases and just 8 days for complaints.⁴ Hence, the memory for prior purchases is long, while complaints are forgotten quickly in the customer's purchasing behavior. Taken together, these results imply that unless the customer leaves the company after the first complaint, or a close second failure occurs, the relationship quickly returns to business-as-usual.

Lastly, we turn to the timing process in Table 1. The Weibull parameter estimates are significantly different from those of an exponential model ($\chi^2_4 = 596.68, p < .01$); hence we

can justify our Weibull timing model on statistical grounds. The estimates indicate that customers in Class 1 purchase almost yearly (356 days) and that the average time until a complaint is voiced (if it is not voiced on the same day as a purchase) is 11 days. Customers in Class 2 purchase more frequently (121 days) and take longer to complain (110 days); the prior complaints variable is usually very close to zero when complaints occur in this class, leading to the earlier identification issue. Complaint incidence (i.e., the probability that a purchase is followed by a complaint rather than a purchase, π_{com}) is about 3% in both latent classes. About 19% of complaints occur on the same day as purchase (π_{sameday}), and 8% of complaints are followed by another complaint rather than a purchase (π_{another}).

The Financial Consequences of Complaints and Recoveries

The previous discussion suggests that customer complaints, prior complaints and purchases and company recovery efforts can have a dramatic impact on customer churn. We now examine the financial consequences of these factors by computing their impact on residual lifetime value (RLV).

There are many possible scenarios to illustrate the financial impact of complaint and recovery and how it interacts with prior complaints and purchases. We consider two customer types: a new customer without any repeat purchases (the majority of customers in our data) and an established customer who made five repeat purchases. After the last purchase either there is no complaint or a complaint that occurs 28 days later. We consider three settings for prior complaints: none, one prior complaint 28 days before the current complaint (i.e., on the day of the most recent purchase), and one prior complaint that occurred only 7 days before the current

complaint. Appendix C provides a detailed description of the various scenarios and the procedure to compute RLV.

First we show the effects in Figure 3 for a new customer with a single purchase. The RLV of this customer is \$108 in the absence of a complaint. If this is followed by a complaint 28 days later, the company can decide to recover the complaint, in which case the RLV is \$100, or the company can withhold recovery, in which case the RLV is \$58. In other words, the value of recovery to the company for a new complaining customer without prior complaints is $\$100 - \$58 = \$42$. Since the value of recovery is larger than a reasonable estimate of its costs, the company should always recover these complaints.

[Insert Figure 3 About Here]

What if this complaint was not the first made by this new customer? If she had made a complaint in the recent past (seven days ago), her RLV drops to \$57 if the company provides recovery for the current complaint and \$39 if no recovery is provided. RLV is quite low regardless of whether recovery is provided; even with recovery, she is likely to churn after the current complaint that occurs close after the first. The value of recovery for this customer is $\$57 - \$39 = \$18$. If the prior complaint was made 28 days before the current complaint, the RLV is \$89 if recovery is provided and \$52 if it is not provided—still less than if no prior complaint had occurred but substantially more than when the prior complaint was seven days ago; the effect of prior complaints diminishes rapidly.

[Insert Figure 4 About Here]

Figure 4 shows the analogous results for an established customer with five repeat purchases. The hassle-free RLV is \$221. If this customer complains 28 days after her last purchase and the company provides a recovery, the RLV is \$217, almost that in the case of no

complaint. If no recovery is provided, the RLV drops to \$179, implying that the value of recovery for an established customer without prior complaints is $\$217 - \$179 = \$38$. If there is a recent prior complaint (7 days before the current complaint), RLV drops to \$166 with recovery and to \$118 without recovery; the value of recovery is $\$166 - \$118 = \$48$. If there is a more distant prior complaint (28 days before the current complaint), RLV is again between that in the cases without any prior complaint and with a complaint seven days ago. Taken together, the value of recovery is slightly higher for new customers (\$42) than for established customers (\$38) when there are no prior complaints, but this pattern is reversed when a prior complaint is present and a second complaint occurs (\$18 and \$48 if the prior complaint was 7 days ago).

Implications for Theory

Our study is the first to integrate complaint management with customer base analysis. In contrast to the literature, we take the company's perspective by studying the effects of actual customer complaints and company recoveries on actual customer purchasing. This is a key point of differentiation; virtually all studies measure recovery effectiveness with survey-based metrics like satisfaction or repurchase intentions (Parasuraman 2006). We tested three fundamental hypotheses from the services literature using actual behavior and a new hypothesis regarding the duration of the effects of prior purchases and complaints on the customer relationship. Our approach and findings have several implications for theory.

Customer churn as a self-selection mechanism. Consistent with H1, longitudinal survey evidence (Maxham and Netemeyer 2002) and attribution theory (Folkes 1984), customer complaints are more harmful if the customer has complained before. But how long do the

negative effects last? Extant theory suggests that negative experiences are more memorable than positive ones (e.g., Baumeister et al. 2001). Yet what we find is the opposite: the effects of complaints are short-lived, with a half-life of just 8 days, much shorter than those of purchases (H4b).

We can reconcile these findings by considering the role of customer churn as a self-selection mechanism (Hausman and Wise 1979). Customers sort themselves following a complaint. Those who are most dissatisfied drop out of the customer base; those who remain are willing to give the company another chance. Sorting effects have been documented in the customer base (e.g., Fader and Hardie 2010) and pricing (e.g., Danaher 2002) literatures, but have received scant attention in the service recovery literature. A key implication is that if all complaints voiced by customers—both those likely to churn and those likely to stay—are considered, the duration of negative effects of complaints on future purchasing are likely to be overstated.

Relationship building by transaction building. According to extant literature, prior satisfaction creates a buffer that mitigates the negative impact of a failure on current satisfaction, trust and purchase intentions (e.g., Tax, Brown and Chandrashekar 1998; Van Doorn and Verhoef 2008). Consistent with this, we find that prior purchases alleviate the effect of a complaint on churn (H2). Importantly and to the best of our knowledge not considered before, we find that the impact of prior purchases is long-lasting, with a halving time of almost one year. While the customer base literature suggests that recency of last purchase is often more crucial than the number of prior purchases (e.g., Kumar and Reinartz 2012), our results suggest that purchase frequency may actually be the most important driver of post-complaint response; this

suggests that the number of prior purchases is a simple but quite good proxy for the strength of the customer relationship (e.g., Hess, Ganesan and Klein 2003).

Measuring recovery effectiveness with customer behavior. Estimates of recovery effectiveness reported in the literature vary widely and range from a null effect (e.g., Bolton 1998; Van Doorn and Verhoef 2008) to the case where recovery more than offsets the loss associated with a complaint, the recovery paradox (e.g., Maxham and Netemeyer 2002; Smith and Bolton 1998). In contrast to these studies, our results are based on actual complaint, recovery and purchase behavior. We show that recovery has a substantial impact on churn (H3), mostly offsetting the loss associated with a complaint, but find little evidence of the recovery paradox. Given the weak link between intentions and behavior (e.g., Gupta and Zeithaml 2006; Chandon, Morwitz and Reinartz 2005), it is important to use behavioral data to estimate recovery effectiveness.

Implications for Managers

Our model and empirical results offer several new implications for managers. We take a first step toward a decision support system using customer base analysis for financially accountable service recovery.

Recovery based on costs and benefits. While the basic concept of using cost-benefit analysis to determine recovery is straightforward and has been around for over 20 years, to the best of our knowledge no paper has put this into practice. For example, Rust, Subramanian and Wells (1992) provide a conceptual illustration in which recovery lifts a repurchase rate from 60% to 90%; they explain that if RLV of a retained customer is \$200, recovery is cost effective until

$(0.90 - 0.60) \times \$200 = \60 . Goodman (2006) offers a similar calculation. Though these back-of-the-envelope calculations are insightful, they require several strong assumptions on RLV and how the customer's churn probability (and hence RLV) is affected by complaints. Another complication is that in non-contractual settings customer churn is unobserved, making it hard to attach numbers to it. Our approach addresses these issues.

Prevent when possible, recover when needed. In our simulation, the value of recovery is often substantially higher than the costs. Recovery is an effective tool to reset the customer relationship: it substantially reduces immediate churn due to the complaint and the delayed effect of the incident is short-lived for customers who remain with the company. Nevertheless, the virtual absence of a recovery paradox for actual post-complaint purchasing makes it worthwhile to devote resources to prevent failures from occurring, because recovery is costly and does not result in a "better" customer compared to the hassle-free case. Though we cannot make precise claims about how much to invest in failure prevention, our results imply that it is better to prevent than recover if prevention is sufficiently cheap.

Do not always prioritize new customers. A widespread belief is that managers should focus recovery resources on new customers, because their preferences are most influenced by negative disconfirmation (e.g., Bolton 1998; Rust et al. 1999). Unlike previous studies, we consider long-term financial impact, and our findings add an important qualifier to this result. Our value of recovery calculations show that it is most profitable to provide recovery to an established customer when there is a recent complaint; such customers may not feel appreciated enough when recovery is refused and the previous incident is still fresh. Furthermore, even without any prior complaint, the value of recovery was only slightly higher for new than for established customers. While for new customers the likelihood of churn is indeed affected most

by the company's recovery decision, the expected future profits at stake are smaller; the value of recovery is determined by both.

Further Research

There are a number of directions to extend this research. One could add behavioral complexity to the model by breaking up complaints and recoveries into specific types or dimensions (e.g., Smith and Bolton 1998; Tax, Brown and Chandrashekar 1998), including interactions between complaint and recovery types (Smith, Bolton and Wagner 1999). Furthermore, our model can be extended by making the purchase timing process dependent on customers' prior purchases and complaints and whether complaints are recovered.⁵ It would also be useful to develop an approach that accounts for word-of-mouth or other network effects across customers to capture the impact of such customer interactions on churn and ultimately RLV (Hogan, Lemon and Libai 2003). Collecting self-reported perceptual measures, such as customer satisfaction, and merging them with longitudinal purchase and complaint data (Gupta et al. 2006; Gupta and Zeithaml 2006) may comprise the backbone of any company's service-quality information system and allow for a more detailed assessment (Berry and Parasuraman 1997).

While we have investigated the consequences for churn, we have ignored the aggregate-level implications for customer equity and company value (Gupta, Lehmann and Stuart 2004; Rust, Lemon and Zeithaml 2004). Companies that have superior recovery capabilities may enjoy greater customer acquisition and are likely to have a greater interaction orientation (Ramani and Knox 2014) and hence have better customer relationships and higher profits (Ramani and Kumar 2008). While there are several good examples of companies with a reputation for superior

recovery capabilities (e.g., Zappos), there is scant research systematically linking recovery and customer equity. Furthermore, while we focus on the consequences of complaints, studying their antecedents within the company, such as product or service failures, or service quality, would shed more light on the relative merits of preventing failures and recovering from failures.

Lastly, from a more methodological point of view, manual coding could be replaced with automated content analysis of the complaint log file to take full advantage of promising innovations in electronic text-mining (Coussement and Van den Poel 2008; Kanaracus 2008). We hope this work will inspire further research on these and other topics.

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Footnotes

1. We extended the churn probability after a complaint with two terms interacting prior complaints and purchases with complaint recovery, but their effects were not significant.
2. $\pi_{\text{churn_com}} < \pi_{\text{churn_pur}}$ implies that $\delta_{\text{com},0} + \delta_{\text{com},1}\text{PCOM} + \delta_{\text{com},2}\text{PPUR} + \delta_{\text{com},3}\text{REC} < \delta_{\text{pur},0} + \delta_{\text{pur},1}\text{PCOM} + \delta_{\text{pur},2}\text{PPUR}$. Setting $\text{PCOM} = 0$ (no prior complaints), $\text{REC} = 1$ (recovery is provided), and rewriting yields $\text{PPUR} > (\delta_{\text{com},0} + \delta_{\text{com},3} - \delta_{\text{pur},0})/(\delta_{\text{pur},2} - \delta_{\text{com},2})$.
3. As mentioned earlier, we estimate this one parameter for both classes for identification.
4. For purchases, the halving time x corresponds to $0.9978^x = 0.5$. By taking logarithms on both sides, it can be written as $x\ln(0.9978) = \ln(0.5)$, or equivalently, $x = \ln(0.5)/\ln(0.9978) = 315$ days. The halving time for complaints is computed similarly.
5. For complaining customers with a non-censored post-complaint interpurchase time we compared its average value for recovered and non-recovered complaints. These averages were 167 and 204 days, respectively, and the difference was not statistically significant. Furthermore, the difference may be driven by company targeting, as the company is somewhat more likely to provide recovery to customers with many prior purchases, i.e., shorter interpurchase times.
6. The empty product runs from $j = 2$ to 1 and occurs if there were no events after the customer's initial purchase.
7. Instead of using the recovery model (8) for future complaints, we also considered a policy of "always recover" or "never recover", based on the company's response to the current complaint, which we manipulate in the various scenarios. The RLV results did not differ by more than 12% and the substantive findings remained unaffected.
8. The RLV results change by less than 2% if the cost of recovery is reduced from \$20 to zero; they do not critically depend on the cost of recovery.
9. We weigh the classes by their posterior probabilities to incorporate all information about unobserved class membership. As prior purchases and complaints play a dual role (i.e., non-stationarity in the customer's churn probability and heterogeneity signalling), we also simulated RLV using prior probabilities, i.e., class sizes. In none of the scenarios RLV changed by more than 10% and the substantive findings are robust.

Table 1
Parameter Estimates (Standard Errors Between Parentheses)

Parameter	Description (Hypothesis)	Class 1	Class 2
κ	Class size	.82 (.01)	.18 (--)
$\delta_{com,0}$	Churn after complaint – intercept	2.33 (.70)	.27 (.44)
$\delta_{com,1}$	Churn after complaint – prior complaints (H1)	2.70 (1.49)	-6.53 (9.87)
$\delta_{com,2}$	Churn after complaint – prior purchases (H2)	-1.38 (.33)	-.49 (.15)
$\delta_{com,3}$	Churn after complaint – recovery (H3)	-1.13 (.56)	-.18 (.42)
$\delta_{pur,0}$	Churn after purchase – intercept	-.22 (.02)	-.74 (.05)
$\delta_{pur,1}$	Churn after purchase – prior complaints	2.41 (1.01)	.91 (.97)
$\delta_{pur,2}$	Churn after purchase – prior purchases	-1.12 (.06)	-.46 (.03)
$\delta_{rec,0}$	Company recovery decision – intercept	2.95 (.67)	2.84 (.83)
$\delta_{rec,1}$	Company recovery decision – prior complaints	-3.20 (1.35)	
$\delta_{rec,2}$	Company recovery decision – prior purchases	.13 (.39)	.40 (.53)
λ_{com}	Decay rate – prior complaints (H4)	.91 (.04)	
λ_{pur}	Decay rate – prior purchases (H4)	1.00 (.00)	
$\beta_{pur,0}$	Purchase timing – intercept	5.88 (.02)	4.76 (.03)
$\beta_{pur,1}$	Purchase timing – last event complaint	-.12 (.09)	.19 (.12)
α_{pur}	Purchase timing – shape parameter	1.01 (.01)	.93 (.01)
$\beta_{com,0}$	Complaint timing – intercept	2.56 (.03)	4.49 (.12)
$\beta_{com,1}$	Complaint timing – last event complaint	.33 (.13)	-.53 (.37)
α_{com}	Complaint timing – shape parameter	1.75 (.08)	.72 (.04)
π_{com}	Complaint incidence	.03 (.00)	.03 (.00)
$\pi_{sameday}$	Complaint voiced same day as purchase	.19 (.01)	
$\pi_{another}$	Another complaint (second or later)	.08 (.01)	

Figure 1
Conceptual Framework

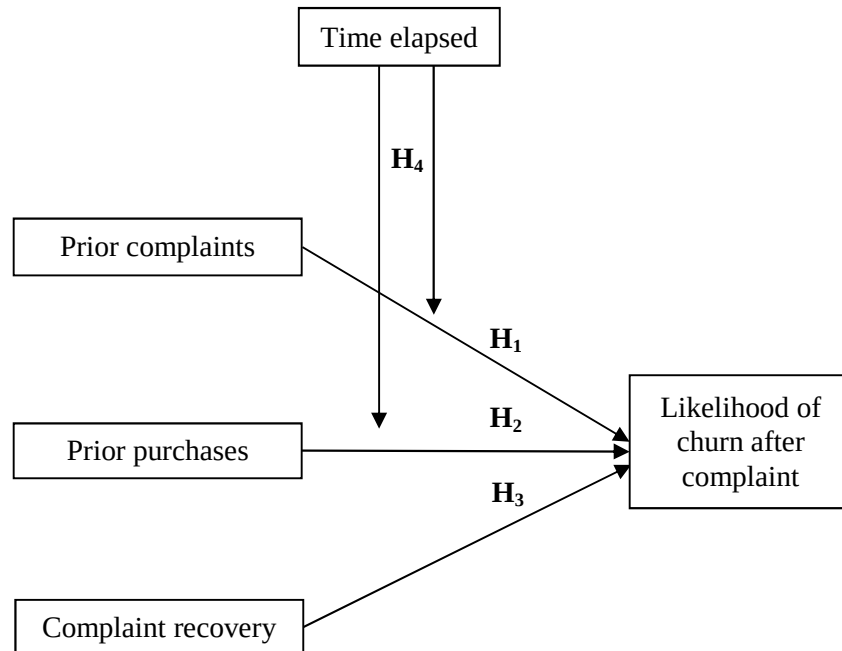


Figure 2

Flow Chart Describing the Customer's Sequence of Steps (Probabilities in Parentheses)

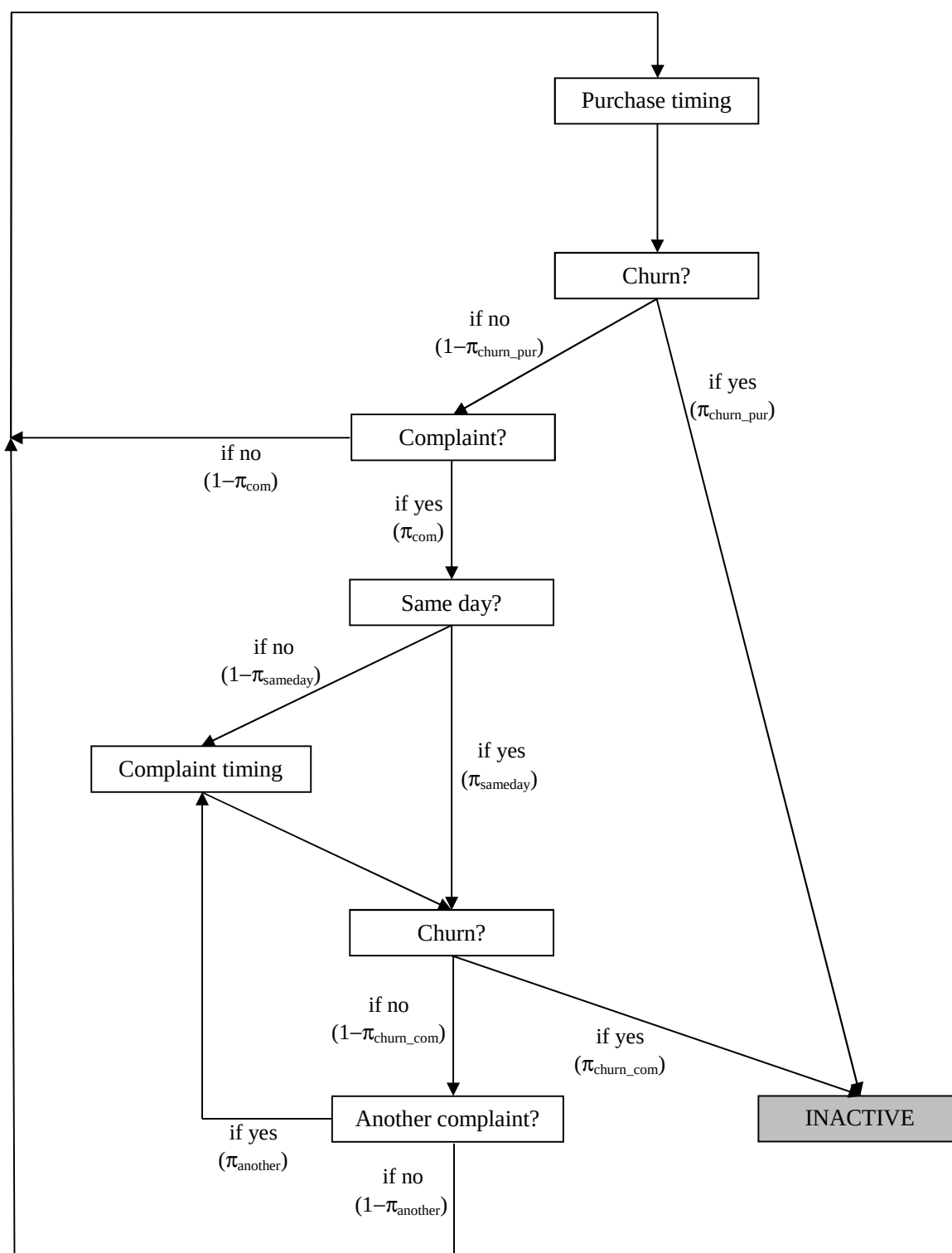


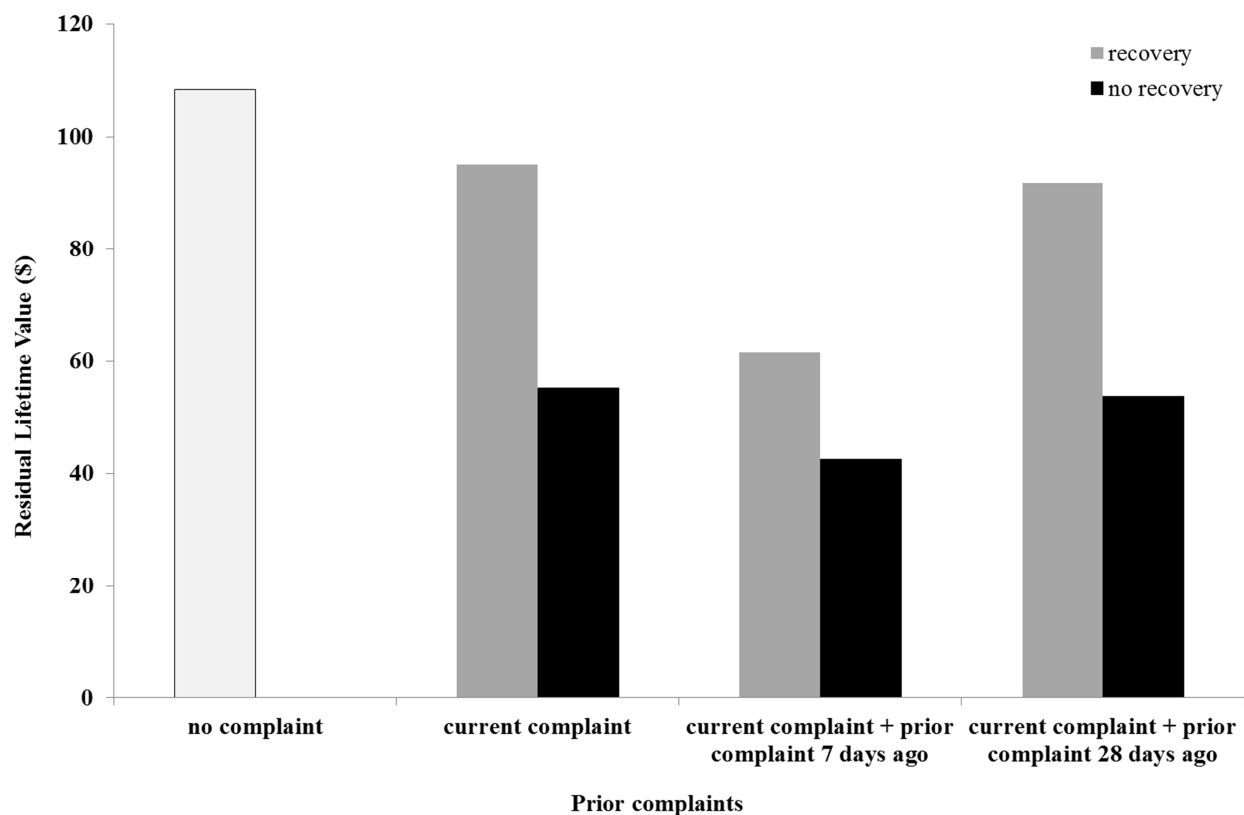
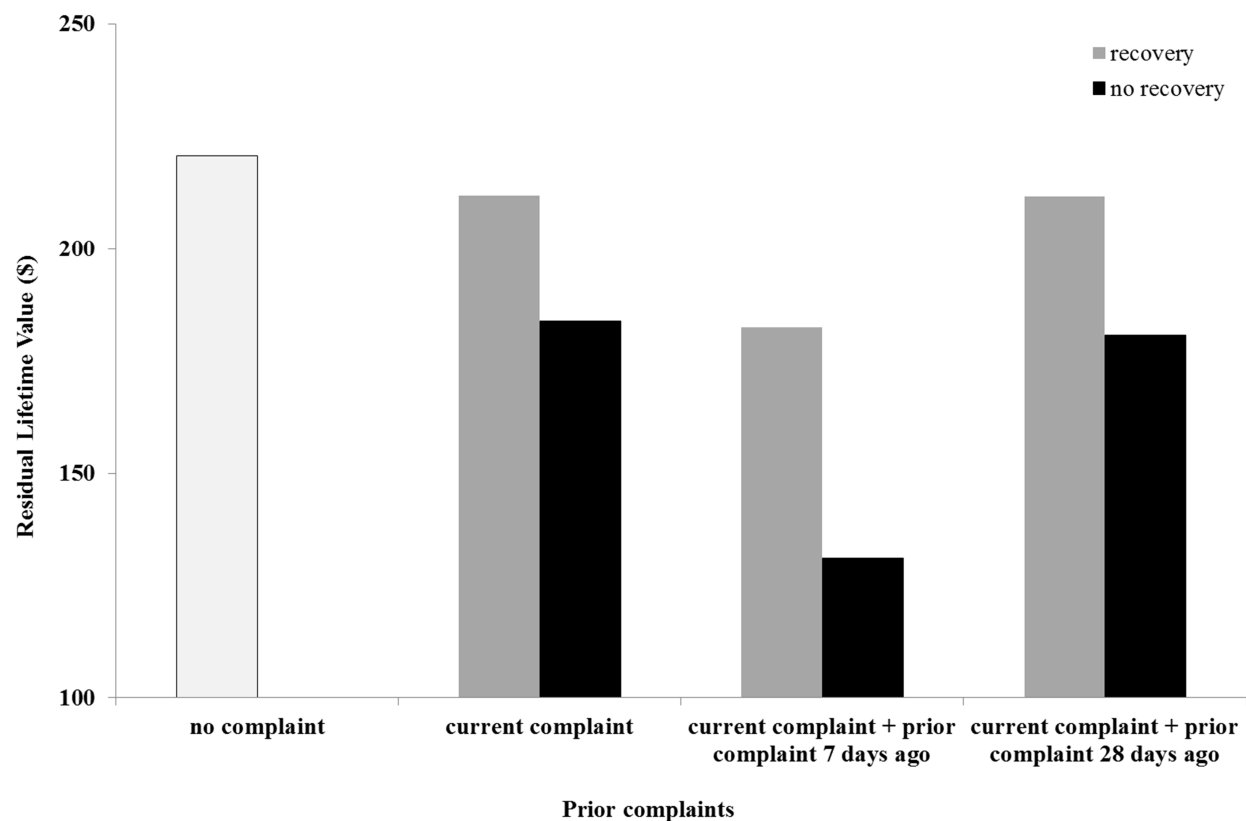
Figure 3**Residual Lifetime Value as a Function of Prior Complaints and Recovery for a New Customer (0 Repeat Purchases)**

Figure 4**Residual Lifetime Value as a Function of Prior Complaints and Recovery for an Established Customer (5 Repeat Purchases)**

Appendix A

This appendix provides the derivation of the (log) likelihood function. We first derive it for one latent class and next extend it to multiple latent classes. We treat all purchases and complaints as separate events, implying that a purchase and a same-day complaint have different indices j . There are three types of events:

	COM _{i,j}	REC _{i,j}
1. Purchase	0	0
2. Complaint without recovery	1	0
3. Complaint with recovery	1	1

Initial Event

For each customer, the initial event $j = 1$ is a purchase with timing $t_{i,1} = 0$. As this event occurs with probability 1, it does not show up in the likelihood.

Repeat Events

Repeat event $j = 2, \dots, J_i$ only occurs if the customer did not churn after the previous event, while also timing and type depend on the last event. If the previous event was a *purchase*, it follows from the flow chart in Figure 2 that the probability of the current event being a purchase with timing $t_{i,j}$ (relative to last event) is given by

$$(A1) \quad (1 - \pi_{\text{churn_pur},i,j-1})(1 - \pi_{\text{com}})f_{\text{pur}}(t_{i,j}),$$

while the probability of a complaint without recovery and with timing $t_{i,j}$ is

(A2)

$$(1 - \pi_{\text{churn_pur},i,j-1})\pi_{\text{com}}\pi_{\text{sameday}}^{I(t_{i,j}=0)}[(1 - \pi_{\text{sameday}})f_{\text{com}}(t_{i,j})]^{1-I(t_{i,j}=0)}(1 - \pi_{\text{rec},i,j}),$$

where the indicator function $I(t_{i,j} = 0)$ equals 1 if $t_{i,j} = 0$ and it is 0 otherwise. Similarly, the probability of a complaint with recovery and timing $t_{i,j}$ is

(A3)

$$(1 - \pi_{\text{churn_pur},i,j-1})\pi_{\text{com}}\pi_{\text{sameday}}^{I(t_{i,j}=0)}[(1 - \pi_{\text{sameday}})f_{\text{com}}(t_{i,j})]^{1-I(t_{i,j}=0)}\pi_{\text{rec},i,j}.$$

If the previous event was a *complaint*, the probabilities become

$$(A4) \quad (1 - \pi_{\text{churn_com},i,j-1})(1 - \pi_{\text{another}})f_{\text{pur}}(t_{i,j}),$$

for a purchase, and

$$(A5) \quad (1 - \pi_{\text{churn_com},i,j-1})\pi_{\text{another}}f_{\text{com}}(t_{i,j})(1 - \pi_{\text{rec},i,j}),$$

$$(1 - \pi_{\text{churn_com},i,j-1})\pi_{\text{another}}f_{\text{com}}(t_{i,j})\pi_{\text{rec},i,j},$$

for a complaint without and with recovery, respectively. Combining (A1)-(A5) yields

(A6)

$$\begin{aligned} L_{\text{repeat},i,j} = & (1 - \pi_{\text{churn_pur},i,j-1})^{1-\text{COM}_{i,j-1}}[(1 - \pi_{\text{com}})f_{\text{pur}}(t_{i,j})]^{(1-\text{COM}_{i,j-1})(1-\text{COM}_{i,j})} \\ & \times [\pi_{\text{com}}\pi_{\text{sameday}}^{I(t_{i,j}=0)}[(1 - \pi_{\text{sameday}})f_{\text{com}}(t_{i,j})]^{1-I(t_{i,j}=0)}]^{(1-\text{COM}_{i,j-1})\text{COM}_{i,j}} \\ & \times [\pi_{\text{rec},i,j}^{\text{REC}_{i,j}}(1 - \pi_{\text{rec},i,j})^{1-\text{REC}_{i,j}}]^{(1-\text{COM}_{i,j-1})\text{COM}_{i,j}} \\ & \times (1 - \pi_{\text{churn_com},i,j-1})^{\text{COM}_{i,j-1}}[(1 - \pi_{\text{another}})f_{\text{pur}}(t_{i,j})]^{\text{COM}_{i,j-1}(1-\text{COM}_{i,j})} \\ & \times [\pi_{\text{another}}f_{\text{com}}(t_{i,j})\pi_{\text{rec},i,j}^{\text{REC}_{i,j}}(1 - \pi_{\text{rec},i,j})^{1-\text{REC}_{i,j}}]^{\text{COM}_{i,j-1}\text{COM}_{i,j}}. \end{aligned}$$

Silent Period after Customer's Last Event

The silent period of length $T_{\text{silent},i}$ between the customer's last observed event and the end of the observation period either implies that the customer became inactive after the last event

or that the next event (purchase or complaint) occurred beyond the observation horizon. If the last event J_i of customer i was a *purchase*, the probability that the customer became inactive is $\pi_{\text{churn_pur},i,J_i}$, while the probabilities of a purchase or a complaint outside the observation period are

$$(1 - \pi_{\text{churn_pur},i,J_i})(1 - \pi_{\text{com}})S_{\text{pur}}(T_{\text{silent},i}),$$

$$(1 - \pi_{\text{churn_pur},i,J_i})\pi_{\text{com}}(1 - \pi_{\text{sameday}})S_{\text{com}}(T_{\text{silent},i}),$$

respectively, where

$$S_{\text{pur}}(T_{\text{silent},i}) = \exp(-[T_{\text{silent},i}/\tilde{\beta}_{\text{pur},i,j}]^{\alpha_{\text{pur}}}),$$

$$S_{\text{com}}(T_{\text{silent},i}) = \exp(-[T_{\text{silent},i}/\tilde{\beta}_{\text{com},i,j}]^{\alpha_{\text{com}}}),$$

are the survival function counterparts of $f_{\text{pur}}(t_{i,j})$ in (1) and $f_{\text{com}}(t_{i,j})$ in (3). Analogously, if the last event was a *complaint*, the probability that the customer became inactive is $\pi_{\text{churn_com},i,J_i}$, while the probabilities of a purchase or a complaint outside the observation period become

$$(1 - \pi_{\text{churn_com},i,J_i})(1 - \pi_{\text{another}})S_{\text{pur}}(T_{\text{silent},i}),$$

$$(1 - \pi_{\text{churn_com},i,J_i})\pi_{\text{another}}S_{\text{com}}(T_{\text{silent},i}),$$

respectively. Combining all possible cases yields

(A7)

$$\begin{aligned} L_{\text{silent},i} = & \{\pi_{\text{churn_pur},i,J_i} + (1 - \pi_{\text{churn_pur},i,J_i})(1 - \pi_{\text{com}})S_{\text{pur}}(T_{\text{silent},i}) \\ & + (1 - \pi_{\text{churn_pur},i,J_i})\pi_{\text{com}}(1 - \pi_{\text{sameday}})S_{\text{com}}(T_{\text{silent},i})\}^{1-\text{COM}_{i,J_i}} \\ & \times \{\pi_{\text{churn_com},i,J_i} + (1 - \pi_{\text{churn_com},i,J_i})(1 - \pi_{\text{another}})S_{\text{pur}}(T_{\text{silent},i}) \\ & + (1 - \pi_{\text{churn_com},i,J_i})\pi_{\text{another}}S_{\text{com}}(T_{\text{silent},i})\}^{\text{COM}_{i,J_i}}. \end{aligned}$$

Likelihood Function (per Customer per Latent Class)

Using (A6) and (A7), the likelihood for customer i is given by

$$(A8) \quad L_i = \left[\prod_{j=2}^{J_i} L_{\text{repeat},i,j} \right] L_{\text{silent},i},$$

where by convention the empty product has value one.⁶

Log-Likelihood Function with Latent Classes

If we denote the value of (A8) resulting from the parameter values in class z by $L_{i|z}$, the log-likelihood function with Z latent classes and N customers is given by

$$(A9) \quad \ln L = \sum_{i=1}^N \ln \left(\sum_{z=1}^Z \kappa_z L_{i|z} \right),$$

where the relative class sizes κ_z sum to one across classes.

Appendix B

This appendix discusses model validation. We compare our model's predictions to two benchmark models. First, we fit a two-class nested "purchase-only" model: complaint incidents and churn after complaints are excluded. Second, we fit the BG/NBD (Fader, Hardie and Lee 2005a), which uses the prior purchase summary statistics recency and frequency to forecast future purchasing. Since both models ignore the role of complaints, they also ignore the role of the company's recovery decision. Furthermore, they ignore the endogeneity correction, which has been shown to diminish predictive performance in holdout samples (Ebbes, Papies and van Heerde 2011). Hence, the reported predictive performance of our model is conservative.

We calibrate all models using the first 104 weeks (i.e., 2 years) and validate them using the remaining 26 weeks (i.e., 6 months). Because there are no closed-form expressions for expected future purchasing in the complaints and purchase-only models, we must simulate future purchases, complaints and churn following these two types of events. If there is a future complaint, we simulate whether the company provides a recovery using the model estimates of equation (8). Each latent class has its own parameter estimates (Table 1); we simulate from each class separately and combine these simulations with the probabilities of class membership to forecast the expected unconditional future purchases. We use 1,000 simulations to forecast future purchasing. The standard error of the forecasts in the complaints model is low, on average .007 purchases.

We judge model fit using two methods. Following the customer-base literature (e.g., Fader et al. 2005a, b; Van Oest and Knox 2011), we divide customers into groups based on how many repeat purchases they made in the calibration period (0, 1, ..., 6+) and compute the

average number of expected purchases in the validation period in the left panel of Figure B.1.

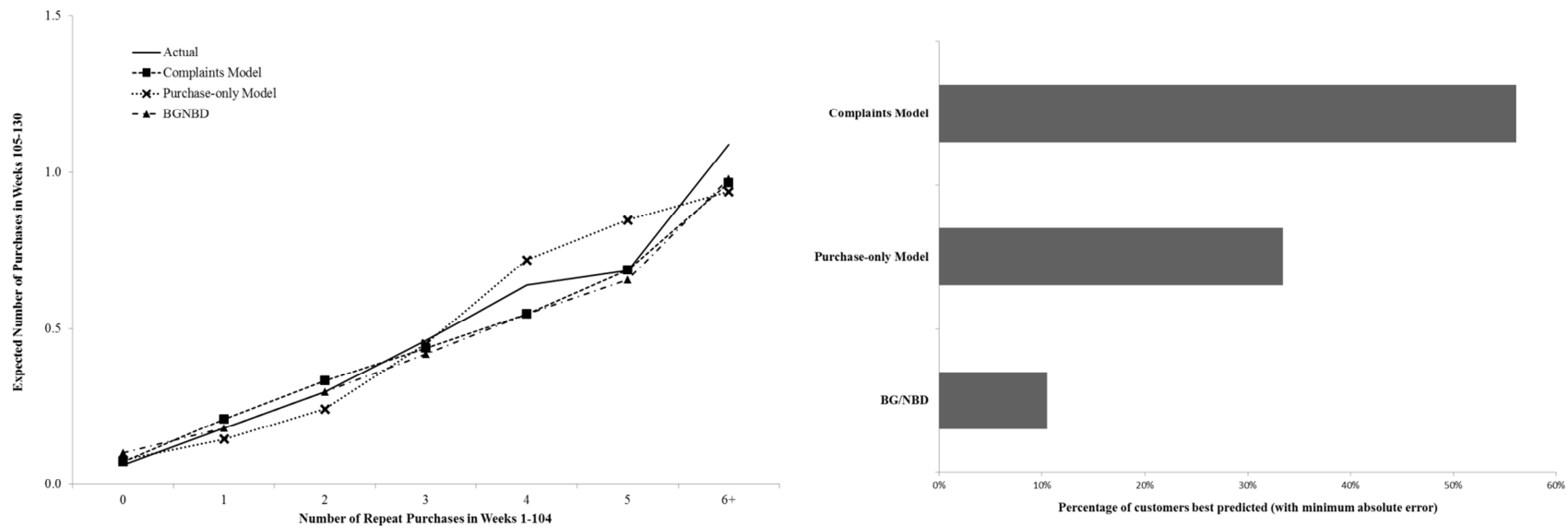
We also count the number of customers best predicted by each model, as measured by the absolute error between the actual number of purchases made by each customer in the validation period and the model predictions in the right panel of Figure B.1.

On the left panel, the grouped predictions figure shows that the predictions of the complaints model (dashed line with squares) and the BG/NBD model (dashed line with triangles) are substantially closer to the actual average number of purchases (the solid line) than the purchase-only model (dashed line with crosses). The absolute error is lower for the complaints model than the purchase-only model in five out of seven cases.

The performance of the complaints model relative to the BG/NBD model is harder to judge. The BG/NBD model is closer to the actual (i.e., has a lower absolute error) in four out of seven cases, but the three cases in which the complaints model performs better account for more total customers than the BG/NBD model. For example, the complaints model has a lower absolute error than the BG/NBD model for the 9,897 customers who made zero repeat purchases in the calibration period, the largest group. Hence, we turn to the right panel in order to see, customer by customer, which model predicts best (in terms of having the lowest absolute error). The right panel shows that the complaints model provides the best predictions for the majority, 56%, of the customer base. Hence, we conclude that the complaints model characterizes future purchasing better than the purchase-only model and the BG/NBD.

Figure B.1

Model Comparison Forecasts of Future Purchasing



Appendix C

This appendix provides the details of the residual lifetime value simulation.

The Scenarios Considered

We always calculate RLV standing immediately after the last event, either a purchase or a complaint. We consider two customer types: a new customer without any repeat purchases (the majority of customers in our data) and an established customer who made five repeat purchases, which we assume split our observation period of 2.5 years evenly, so that each inter-purchase time is 182 days. If there is no complaint (the hassle-free case), the last event is a purchase and we calculate RLV standing immediately after the last purchase. If the last event is a complaint, we assume it occurs 28 days—the empirical average time until a complaint is voiced—after the last purchase; we calculate RLV assuming that we stand immediately after this last complaint. If the last event is a complaint, it may be the only complaint voiced (no prior complaints from this customer), or there may have been a prior complaint voiced 28 days before the current complaint (i.e., on the day of the most recent purchase), or there may have been a prior complaint that occurred only 7 days before the current complaint.

Procedure to Compute Residual Lifetime Value

Fader, Hardie and Lee (2005b) derive an expression for residual lifetime value (RLV) of an already acquired customer, conditional on observed behavior, by combining a model for the inter-purchase time and churn processes with a separate model for profit per transaction. We follow the same path by simulating future purchases, complaints and churn following these two

types of events until the inactive state is reached, i.e., for the entire residual lifetime of each customer. In the process, we keep track of (simulated) prior purchases and complaints. If there is a complaint, we simulate whether the company provides a recovery using the model estimates of parameters $\delta_{\text{rec},0}$, $\delta_{\text{rec},1}$ and $\delta_{\text{rec},2}$ in Table 1.⁷ We impute the company's cost of providing recovery to a future complaint at \$20.⁸ We arrive at this cost estimate by considering a lost shipment: the average order size is \$57.32 and the cost price of the shipped goods is 26% of this amount, i.e., about \$15; we assume the reshipment itself costs about \$5, leading to \$20 in total. Similar to Schweidel and Knox (2013), we simulate separately for each latent class and combine these estimates with the probabilities of class membership.⁹ We simulate purchase amount from a gamma-gamma distribution (Colombo and Jiang 1999; Fader, Hardie and Lee 2005b). The annual discount rate is 10%.

Parameters and Fit of Gamma-Gamma Model for Purchase Amounts

The maximum likelihood estimates of the gamma-gamma distribution (following the notation in Fader, Hardie and Lee 2005b) are $\hat{p} = 2.67$, $\hat{q} = 3.85$, $\hat{\gamma} = 58.52$. The model prediction of the overall mean purchase amount is \$54.82, close to the actual overall mean of \$57.32. We visualize the fit of the gamma-gamma distribution by comparing it to the actual distribution (using kernel smoothing) in Figure C.1. Overall the two distributions are very close. The actual mode, at \$25, is only slightly smaller than the gamma-gamma mode at \$26.

Figure C.1
Gamma-Gamma Model Fit

